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Lab #24:

Linear Search:

```
J linears.java > ...
1 public class linears {
  Run | Debug | Run main | Debug main
2 public static void main(String arg[]){
3   int[] A = {3,4,5,2,1,5,6,2,9,23,1,5};
4   int index = linearSearch(A, key: 5);
5   if(index != -1)
6     System.out.println("Found at "+index);
7   else
8     System.out.println("Not found");
9   }//main
10 private static int linearSearch(int[] A, int key) {
11   for(int i=0; i<A.length; i++)
12     if(A[i]==key)
13       return i;
14   return -1;
15   }
16 }
```

OUTPUT :

```
● owais-hashim@ow
is-hashim/.conf
Found at 2
```

Binary Search :

```
J Bsearch.java > Language Support for Java(TM) by Red Hat > Bsearch
1 public class Bsearch {
  Run | Debug | Run main | Debug main
2   public static void main(String arg[]){
3     int[] A = {1,2,4,5,7,8,9,12,14,16,23,26,27,34,36,38,45,67,88};
4     int index = binarySearch(A, key: 14);
5     if(index != -1)
6       System.out.println("Found at "+index);
7     else
8       System.out.println("Not found");
9     }//main
10 private static int binarySearch(int[] A, int key) {
11   int lowerBound = 0;
12   int upperBound = A.length - 1;
13   int curIn;
14   while (true) {
15     curIn = (lowerBound + upperBound) / 2;
16     if (A[curIn] == key) {
17       return curIn;
18     } else if (lowerBound > upperBound) {
19       return -1 ;
20     } else {
21       if (A[curIn] < key) {
22         lowerBound = curIn + 1;
23       } else {
24         upperBound = curIn - 1; } }
25   }
26 }
27 }
```

OUTPUT :

```
● owais-hashim@owais-  
is-hashim/.config/  
Found at 8
```

Binary Search (Recursive):

```
J binsarysearchrec.java > Language Support for Java(TM) by Red Hat > binsarysearchrec > binarySearchRec(in  
1 public class binsarysearchrec {  
    Run | Debug | Run main | Debug main  
2     public static void main(String arg[]) {  
3         int[] A = {1, 2, 4, 5, 7, 8, 9, 12, 26, 27, 34, 36, 38, 45, 67, 88};  
4         int index = binarySearchRec(A, key:7, lowerBound: 0, A.length);  
5         if (index != -1) {  
6             System.out.println("Found at " + index);  
7         } else {  
8             System.out.println("Not found");  
9         }  
10    } //main  
11    private static int binarySearchRec(int[] A, int key, int lowerBound, int  
12    upperBound) {  
13        int curIn;  
14        curIn = (lowerBound + upperBound) / 2;  
15        if (A[curIn] == key) {  
16            return curIn;  
17        } else if (lowerBound > upperBound) {  
18            return -1;  
19        } else {  
20            if (A[curIn] < key) {  
21                return binarySearchRec(A, key, curIn + 1, upperBound);  
22            } else {  
23                return binarySearchRec(A, key, lowerBound, curIn - 1);  
24            }  
25        }  
26    }  
}
```

OUTPUT :

```
● owais-hashim@owais-hash  
is-hashim/.config/Code/  
Found at 4
```